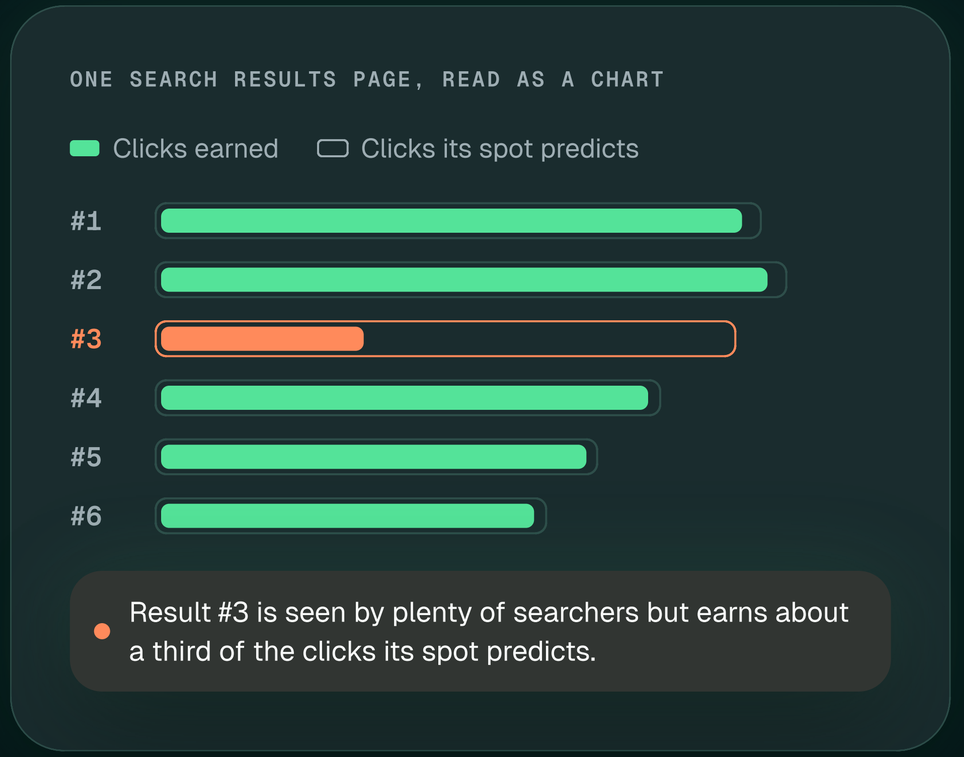
FLYRANK ML INTERNSHIP · LANE 4 · OCTOBER 2026

Which pages keep losing the clicks they should be getting?

A tested way to find pages that rank well on Google but get far fewer clicks than they should, built from 78.8 million daily Google Search records across about 50 websites.



Neil Ivan Orencia
Machine Learning Intern, FlyRank

AT A GLANCE

Lost clicks tend to persist, and a simple rule finds them as well as a machine learning model.

4×

Problems stick around

A page with a real click shortfall this month was 3.8 to 4.8 times as likely to still have one next month.

47 of 50

Both approaches find them

On a month kept untouched until the very end, the top 50 picks held 46 to 47 real shortfalls. Random picks would find about one.

94%

The list holds up

Of the listed pages that nobody touched, 94% were still losing clicks a month later.

RECOMMENDATION Each month, review the 50 pages missing the most clicks, each labelled with the model's confidence and one suggested first step.

UNDER THE HOOD Sealed test month May → June 2026: 137,275 pages, 2.2% with a real shortfall. Top-50 precision 0.94 (logistic regression) vs 0.92 (rule). Persistence measured on pages with 1,000 to 9,999 impressions.

CONTENTS

How this report is organised

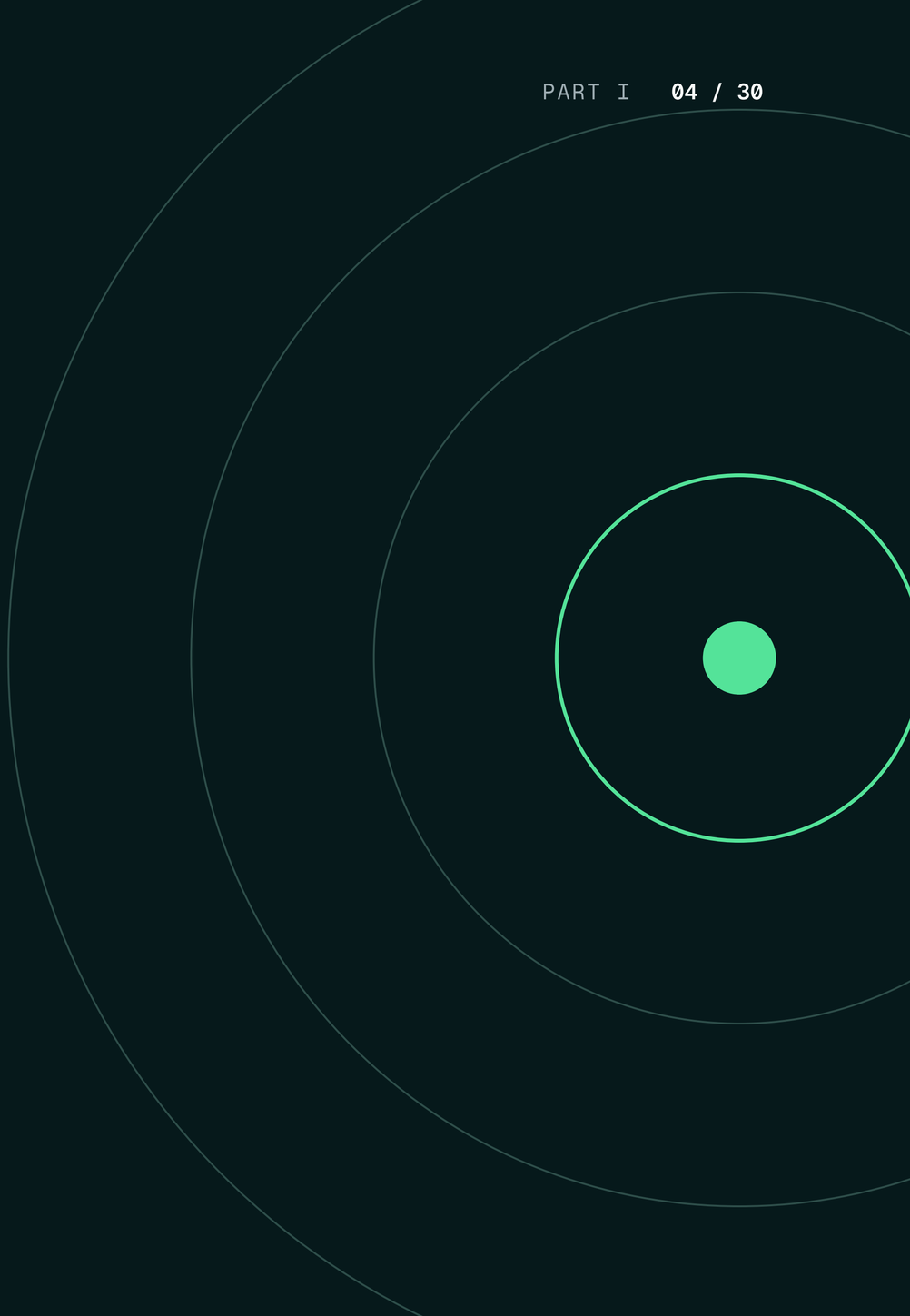
I	The problem	Why a page's clicks can't be judged with one fixed number	04-07
II	The method	How expected clicks, the luck test and a fair test were built	08-16
III	The evidence	What held up, on websites and a month the work had never seen	17-21
IV	The action	A monthly list of 50 pages, each with a suggested first step	22-27
V	Limits and next steps	What this work does not show yet, and what to test next	28-30

PART I

The problem

Search Console shows every page's clicks. The hard part is knowing which pages should be getting more.

05 The editor's problem
06 What the click curve shows
07 Why a flat rule fails



THE DECISION

Editors have thousands of live pages and time to review a few dozen of them each month.

01

Visibility already earned

A page that ranks well but is rarely clicked has done the hard part. A better title, description or page can turn its views into visits.

02

Not every dip is real

Clicks swing a lot by chance. A page can miss clicks for one month and recover on its own, and reviewing it wastes the editor's time.

?

The research question

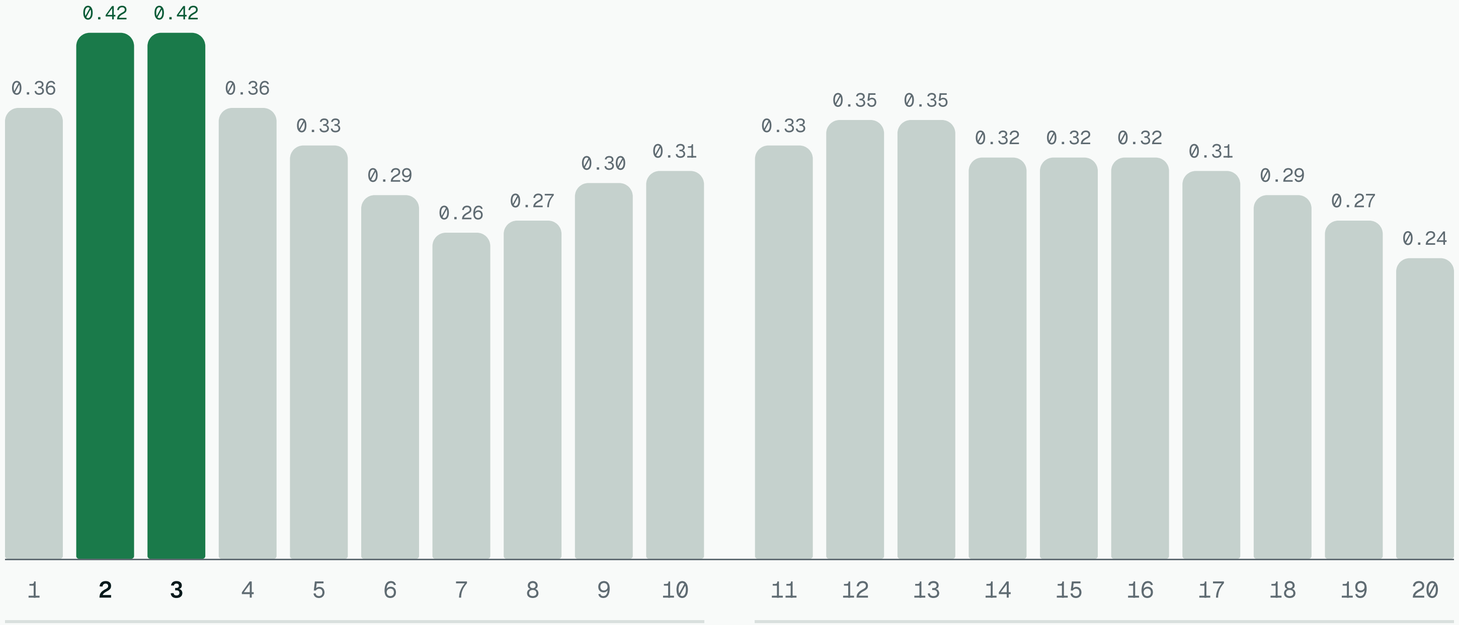
Which pages get far fewer clicks than their spot on Google and their own website predict, beyond bad luck, and will that still be true next month?

UNDER THE HOOD

Decision supported: where to look first in a monthly content review. One row = one page in one calendar-month pair: signals from month M, outcome in month M+1.

WHAT THE DATA SHOWS

Ranking first doesn't guarantee the most clicks. Here, spots 2 and 3 were clicked more often.



0.24% – 0.42%

The normal click-through rate barely changes from position 1 to 20. The top of page 2 (spots 11 to 13) is clicked more than the bottom of page 1 (spots 7 and 8).

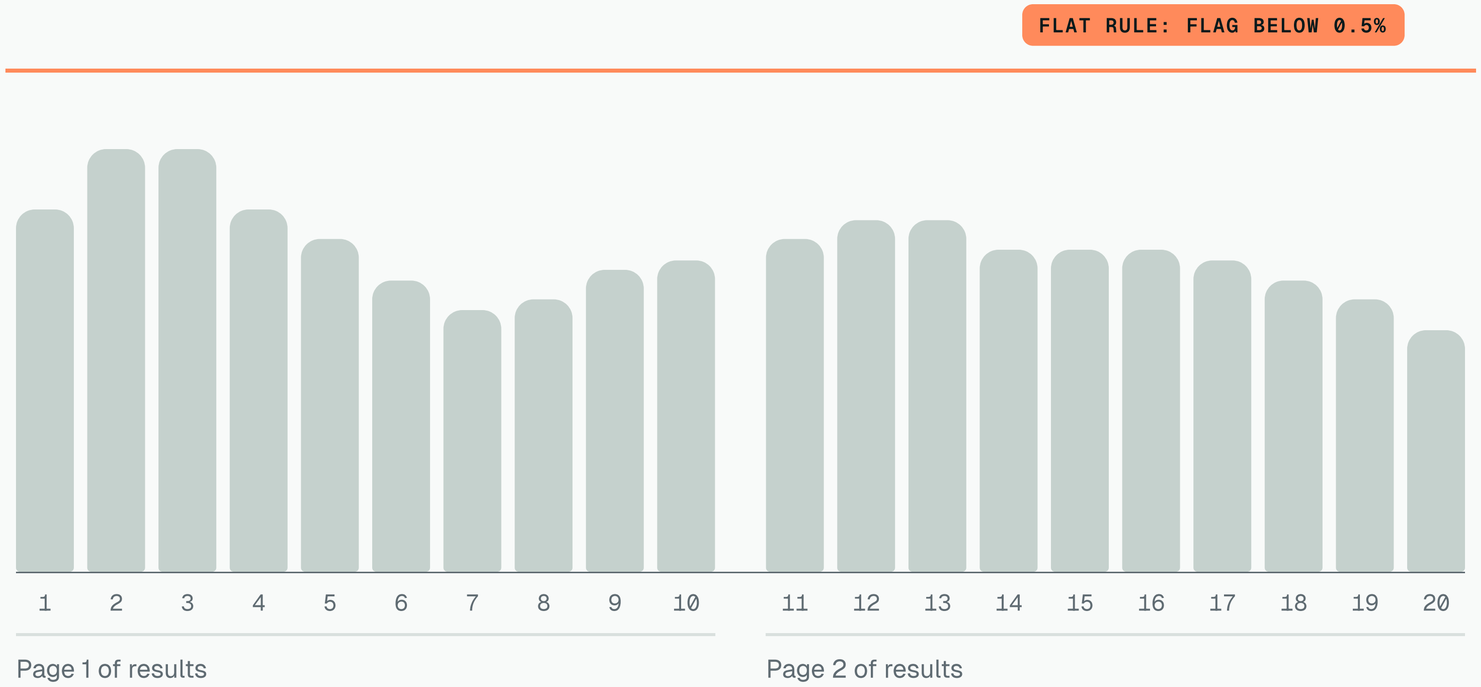
So a page's position alone says little about how many clicks it should get.

UNDER THE HOOD

Normal click-through = pooled clicks ÷ impressions at each position, from the signal audit. Each bar rests on 1,896 to 36,726 pages. Positions below 1 were removed as invalid.

WHY THE OBVIOUS RULE FAILS

A one-size-fits-all rule flags most pages, including the healthy ones.



74% – 88%

of visible pages flagged at every position

Normal click-through sits below the 0.5% line everywhere, so the rule flags pages that are clicking exactly as expected. Each page needs its own yardstick: its spot, its own website, and a test for luck.

UNDER THE HOOD

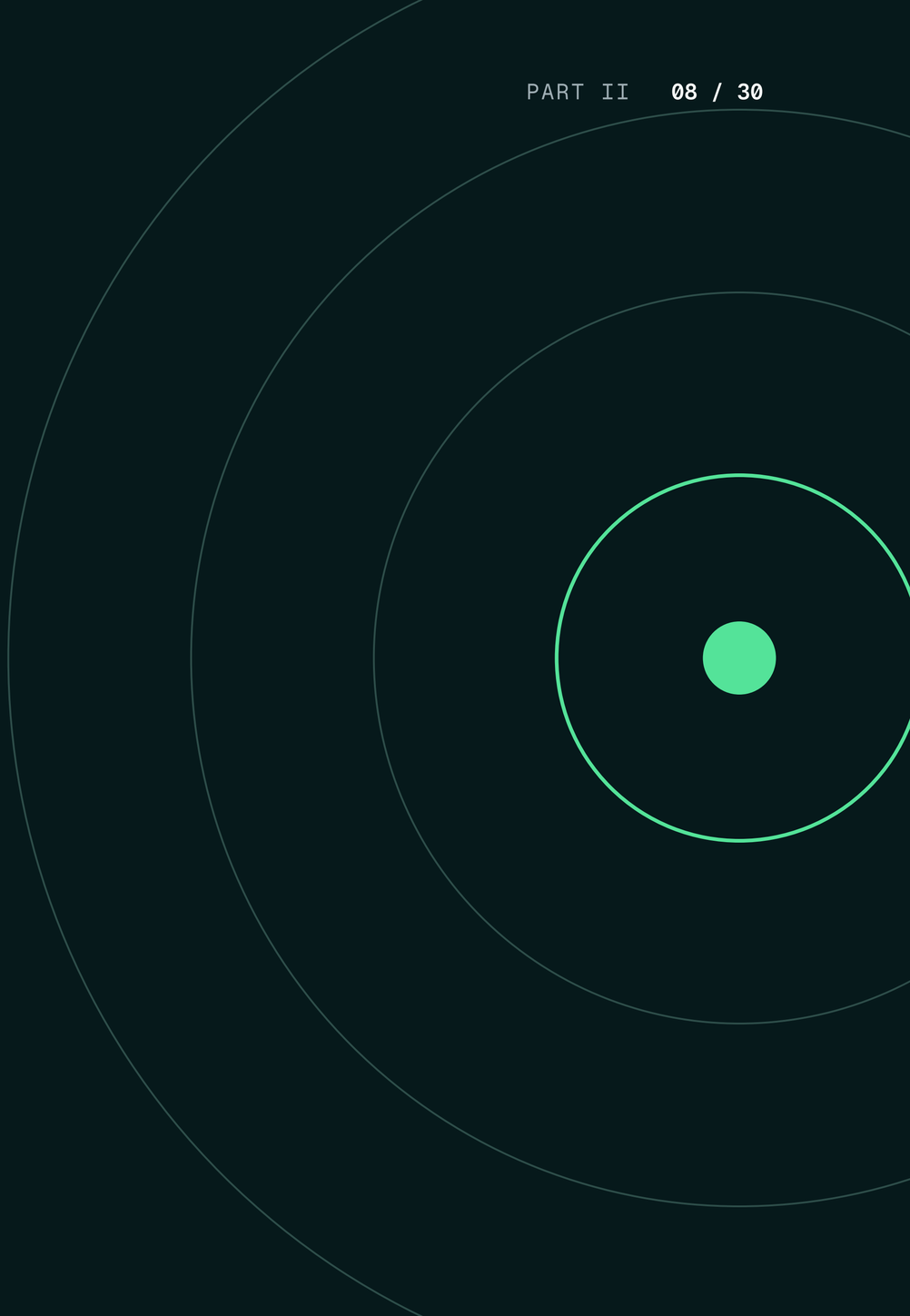
Flat rule tested: ≥ 500 impressions, position ≤ 20, CTR < 0.5%. Share of pages with 500+ impressions flagged per position: 74% (position 2) to 88% (position 20). Websites also differ: in the same spot, some are clicked about 4× as often as others.

PART II

The method

Judge every page against its own spot and its own website, then rule out bad luck before calling it a problem.

09 The data 10 What was left out 11 Expected clicks 12 Clicks are noisy
13 A real shortfall 14 Two contenders 15 A fair test 16 Leakage checks



THE DATA

78.8 million daily records, narrowed to one clean row per page per month.

78,835,655
daily page records

One row per page per day: impressions, clicks, position, and engagement from GA4.

519,606
pages in the content table

Live pages across about 50 client websites, with keyword research for each.

137,275
pages scored on the sealed month

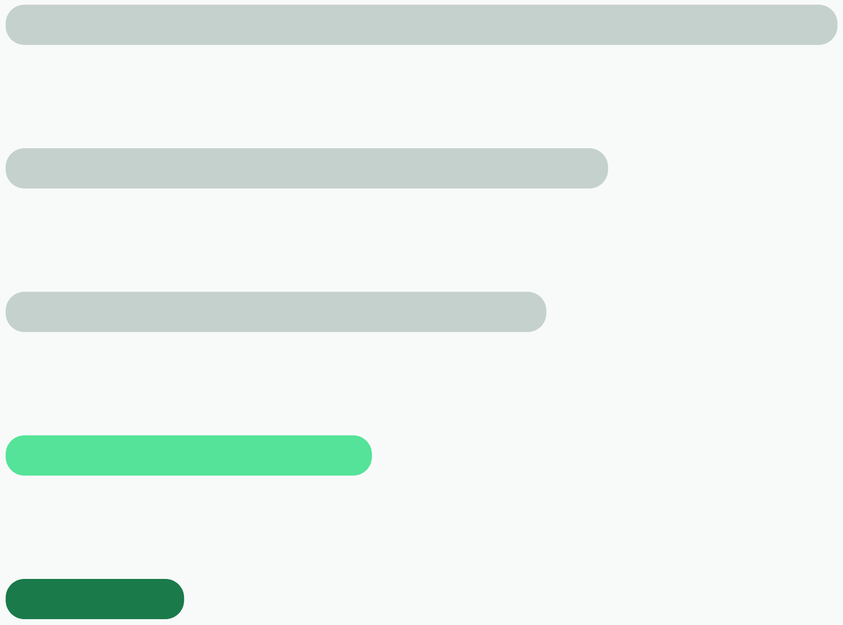
Pages with impressions and a valid position in both May and June 2026.

2,978
real shortfalls in June

2.2% of scored pages. These are what a good list should find.

50
pages on the review list

What one editor can realistically review in a month.



Bar lengths use a log scale, so each step is a big cut.

UNDER THE HOOD

FlyRank internship warehouse, release v20260703, read with DuckDB from Hugging Face. Tables: daily page performance (Search Console + GA4), page content, clients (104), and 90-day queries used only for diagnosis. All identifiers are pseudonymous hashes.

WHAT WAS LEFT OUT

Five kinds of data were left out on purpose, so every comparison is fair.

01	**Impossible positions**	163,189 March page-days showed position 0 and 101,548 a position below 1, which made pages look top-ranked. Valid positions start at 1.
02	**Websites with partial history**	A website whose search history starts mid-month is left out, so every site is measured over a whole month.
03	**Pages missing in either month**	A page needs impressions and a valid position in both months of a pair to be judged.
04	**Pages FlyRank already optimised**	Someone already acted on them during the pair, so next month cannot judge them fairly.
05	**Fields recorded later**	Word counts, edit dates and optimisation flags come from later snapshots, so using them would leak the future.

UNDER THE HOOD

Exclusions are applied per month pair before labelling. Only keyword-research fields that are fixed before a page is written are used as features from the content table.

THE YARDSTICK

Each page is measured against its own spot on Google and its own website.

01

Impressions

How many times the page appeared in Google that month

10,000

views this month

×

02

Normal click-through for its spot

The pooled rate for pages at the same position

0.42%

position 2

×

03

Its website's factor

How much more or less this site is clicked than normal

1.2

site clicked 20% above normal

=

04

Expected clicks

What the page should have earned

≈ 50

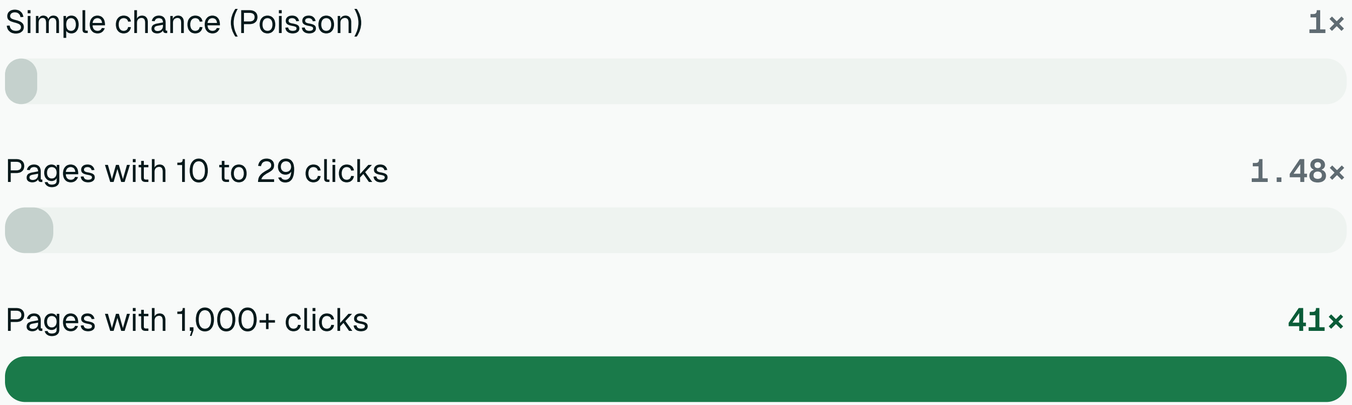
the yardstick

THEN COMPARE If the page actually earned 18 clicks, it got 36% of what it should, about 32 short. That is under half, so it passes the first gate of a real shortfall.

UNDER THE HOOD Position bins: each position 1-10, then 11-20, 21-50 and 51+. Website factor = (site clicks + 1) ÷ (site expected clicks + 1), so a site that is clicked more than normal raises the bar for its own pages.

WHY A LUCK TEST

Clicks swing far more than simple chance allows, and big pages swing the most.



How much clicks vary, compared with simple chance

A test that assumed simple chance would call ordinary swings on big pages "problems".

So the luck test uses a negative binomial model, which allows for the extra swing, with one spread factor fixed before testing.

α = **0.039**

UNDER THE HOOD

Variance relative to a Poisson count. Spread factor α estimated by the method of moments on pages whose position barely moved, once per development pair (0.050, 0.032, 0.039); the median, 0.039, was locked before testing.

THE LABEL

A page counts as a real shortfall only when it passes two gates.

GATE 1

Big enough

The page earned at most half of the clicks it was expected to earn.



actual ≤ ½ × expected

AND

GATE 2

Not bad luck

A luck test says chance explains a gap this big less than 5% of the time.



p < 0.05 under the negative binomial

=

RESULT

Real shortfall

Big enough to matter, and too big to be luck.

UNDER THE HOOD

Target = a real shortfall in month M+1. Share of pages with one: 7.4%, 6.4% and 4.1% in the three development pairs, 2.2% in the sealed pair (May → June 2026).

TWO CONTENDERS

Two ways to rank the pages: a transparent rule and a machine learning model.

RULE B

Rank by missed clicks

Last month's missed clicks (expected minus actual), counted only for pages that were already short last month.

- No training needed
- Easy to explain to an editor
- Naturally favours the biggest click pools

MODEL

Logistic regression

Estimates the chance that a page will have a real shortfall next month, from 25 signals.

- Traffic: impressions, clicks, click-through, position and its daily spread
- Rule B's two signals
- Keyword research: search volume, competition, cost per click, intent, content type

UNDER THE HOOD

Also trained the same way for comparison: a depth-3 decision tree, a random forest and gradient boosting (slide 19). Rule B was chosen from three candidate rules by a criterion fixed in advance, then frozen.

A FAIR TEST

Every test used websites and months the model had never seen, plus one month kept sealed until the end.

MONTH PAIR	Feb → Mar	Mar → Apr	Apr → May	May → Jun
Test 1	Train	Test		
Test 2	Train	Train	Test	
Final test	Train	Train	Train	Sealed · scored once

Unseen websites

Test pages always come from websites left out of that model's training.

Rules fixed in advance

The model wins only if its top-50 hit rate is more than 0.02 higher, or within 0.02 and it captures more missed clicks.

UNDER THE HOOD

Walk-forward and grouped by client. Rows per pair: 114,972 · 155,707 · 157,530 · 137,275. Uncertainty from 1,000 client-level bootstrap resamples. The sealed month was scored once.

LEAKAGE CHECKS

Checks confirmed the model could not peek at next month's answer.

Scrambling next month changes nothing

Shuffling every next-month column, the target and FlyRank's own flags leaves all 25 signals unchanged.

A planted leak gets caught

Adding a deliberate leak lifts the score sharply, which shows the checks can see one.

No test page in training

Zero test pages appear in their model's training rows.

Looser tests would have flattered the model

Top-50 hit rate under each test design

Random rows	0.92
Same months, new websites	0.91
Later months, same websites	0.91
Later months, new websites USED	**0.89**

UNDER THE HOOD

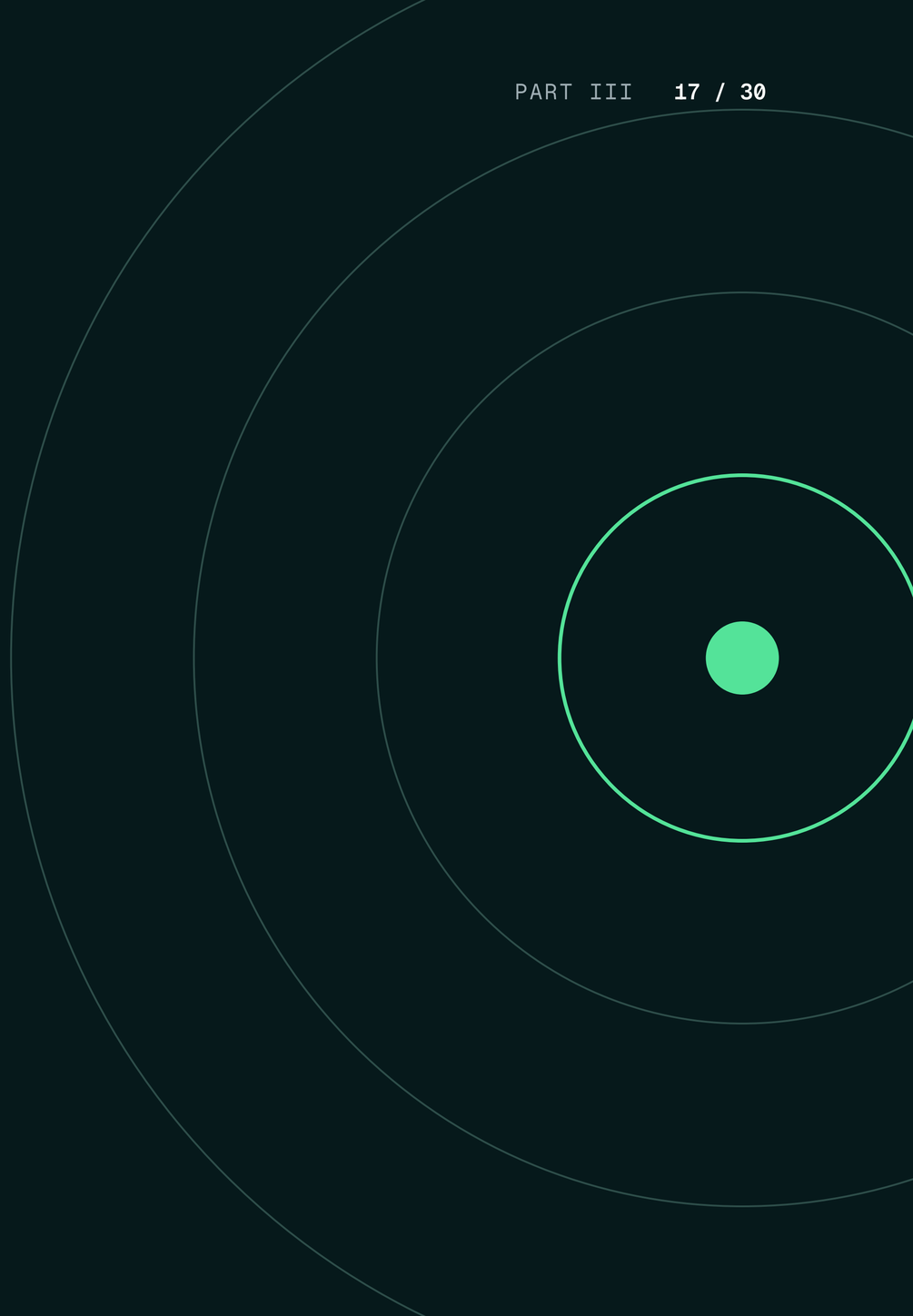
Separation alarm at 0.90: no feature crosses it once page size is held fixed. Leakage scope: every month M+1 column, the target and every product field. Means over the two development test pairs.

PART III

The evidence

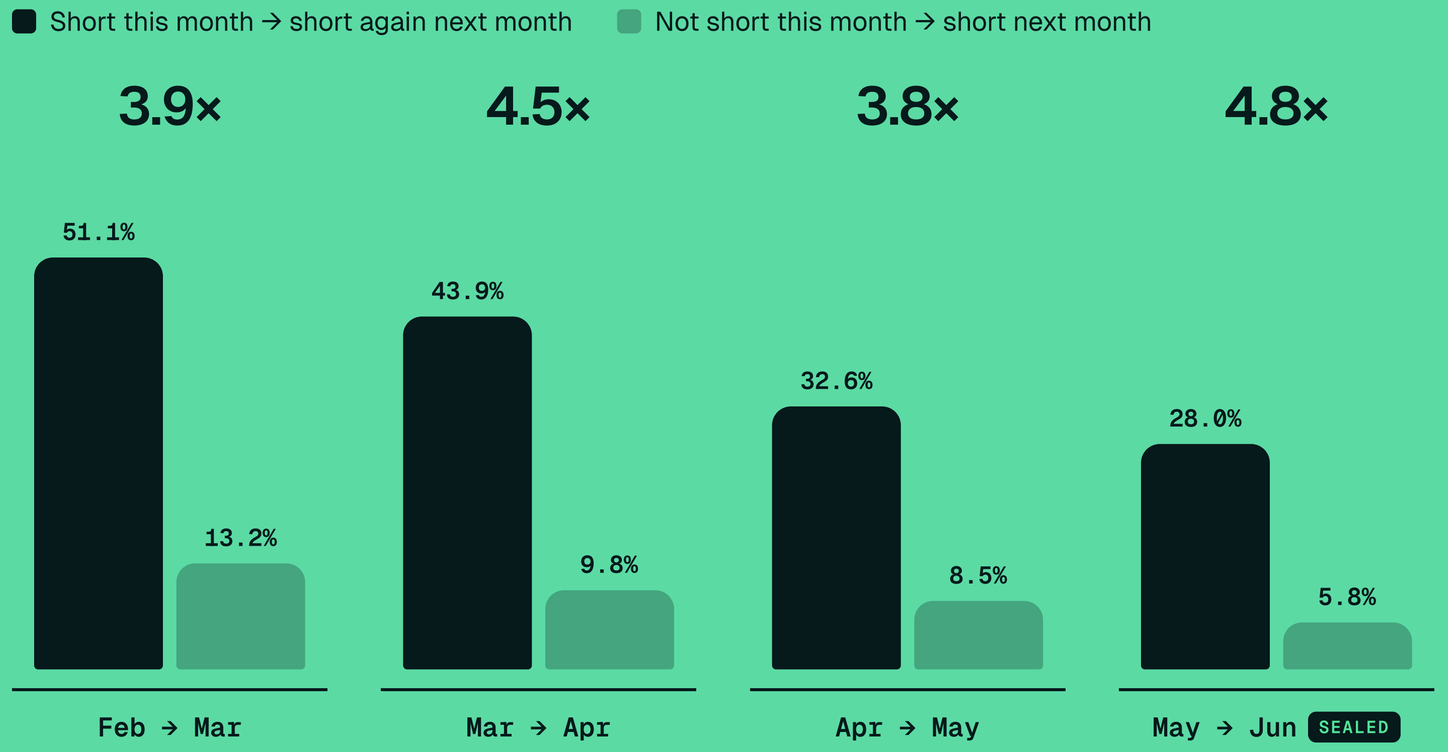
Tested on websites and months the work had never seen, and once more on a month kept sealed until the very end.

18 Shortfalls persist
19 Five ways to rank
20 The sealed month
21 How certain



FINDING 1

Real shortfalls persist: a page with one this month was about 4× as likely to have one next month.



3.8× – 4.8×

as likely to repeat, in every month pair

This is what makes a monthly list worth having: a page flagged this month is usually still worth fixing next month, not a one-off dip.

UNDER THE HOOD

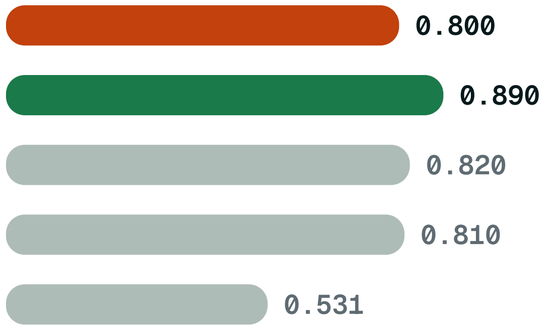
Pages with 1,000 to 9,999 impressions. Pages short in month M: 6,472 · 8,662 · 6,891 · 2,857. Pages not short: 21,259 · 30,355 · 22,779 · 14,063. Holds on the sealed month too.

DEVELOPMENT RESULTS

On the development months, the model led on hits, but the rule's picks held more of the missed clicks.

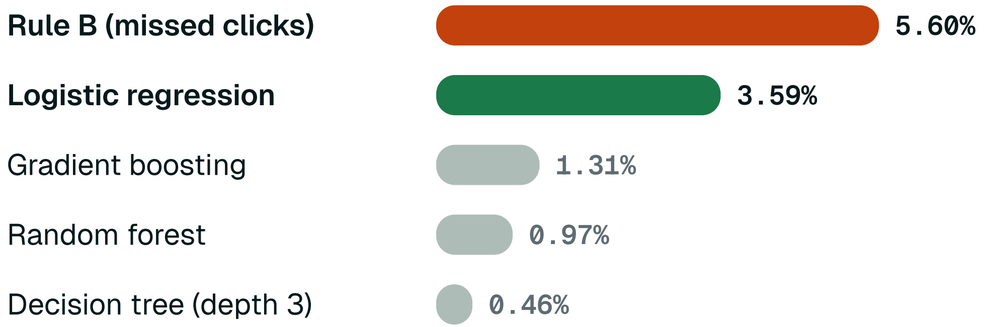
Hits in the top 50

Share of the top 50 picks that had a real shortfall next month (higher is better)



Missed clicks captured

Share of next month's missed clicks held by the top 50 picks (higher is better)

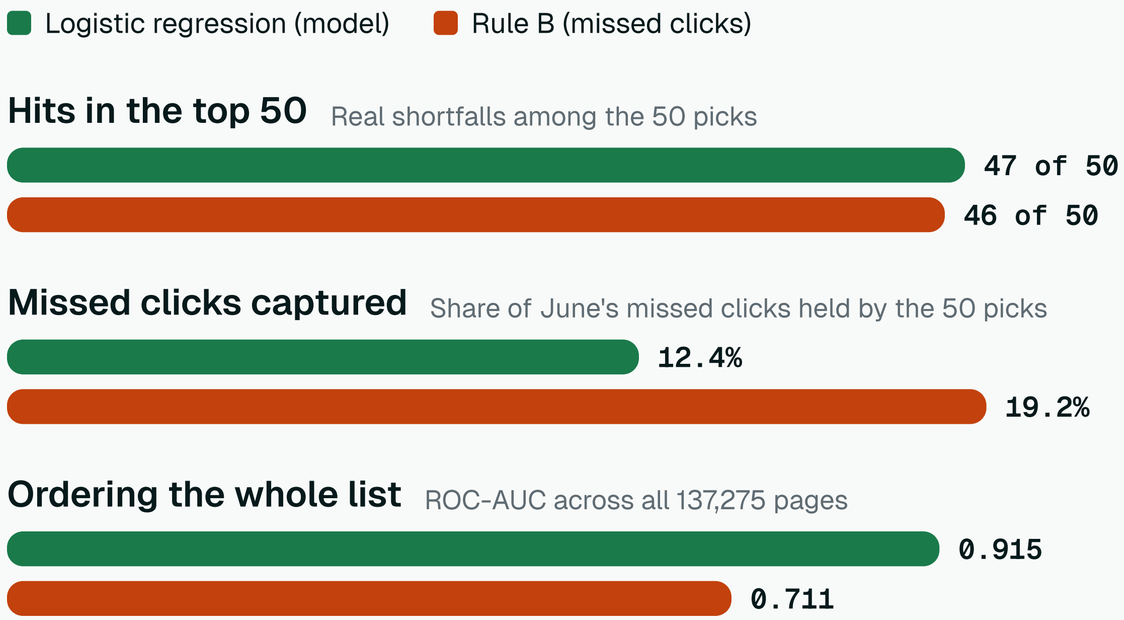


UNDER THE HOOD

Means over the two development test pairs (Mar → Apr, Apr → May). Mean average precision: rule 0.385, logistic regression 0.509, tree 0.415, random forest 0.521, gradient boosting 0.511. Logistic regression was carried forward to the sealed test.

FINDING 2 · THE SEALED MONTH

On the sealed month, both found real shortfalls at nearly the same rate. The rule found bigger click pools.



VERDICT

By the rule fixed in advance, the model did not beat the simple rule.

The 0.02 gap in hits sits inside the margin set before testing, so missed clicks captured decided, and the rule won it. Random picks would find about 1 real shortfall in 50.

UNDER THE HOOD

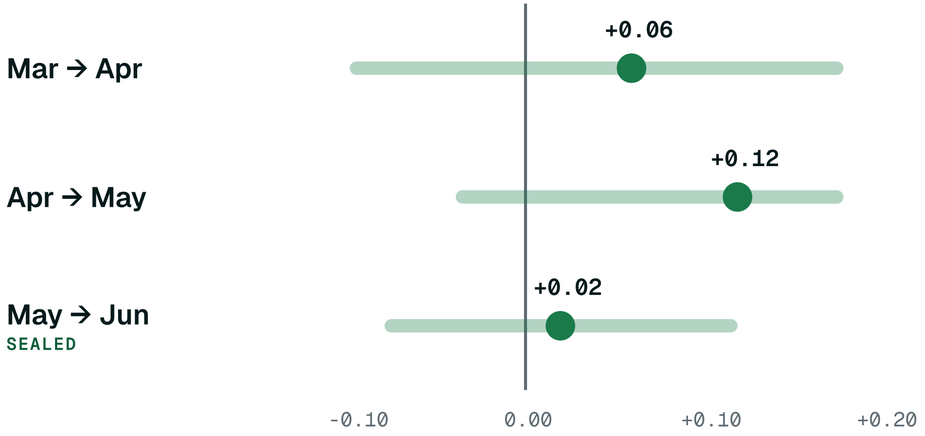
Sealed pair May → June 2026: 137,275 pages, 2,978 real shortfalls (base rate 2.2%). Model trained only on other websites. Average precision: model 0.375, rule 0.294. Top-20 hit rate: model 1.00, rule 0.90.

HOW CERTAIN

The model's lead in hits was never clear of chance. The rule's lead in missed clicks was.

Gap in hits (top 50)

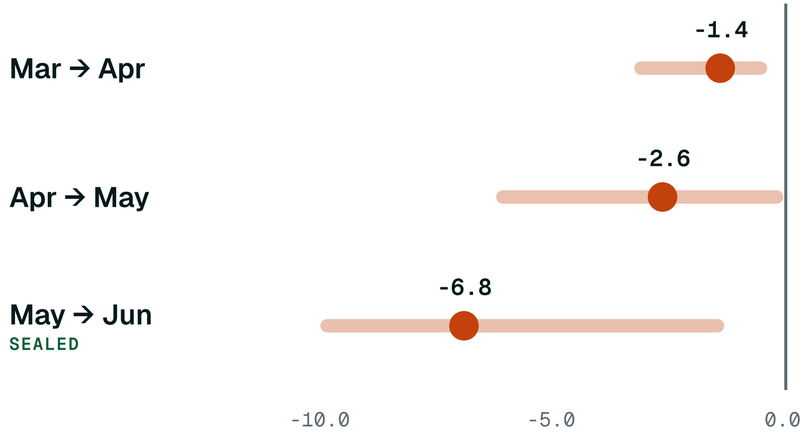
Model minus rule; right of the line = model ahead



Every interval crosses zero: the model's lead could be chance.

Gap in missed clicks captured

Model minus rule, in points; left of the line = rule ahead



Every interval sits left of zero: the rule reliably finds bigger click pools.

UNDER THE HOOD

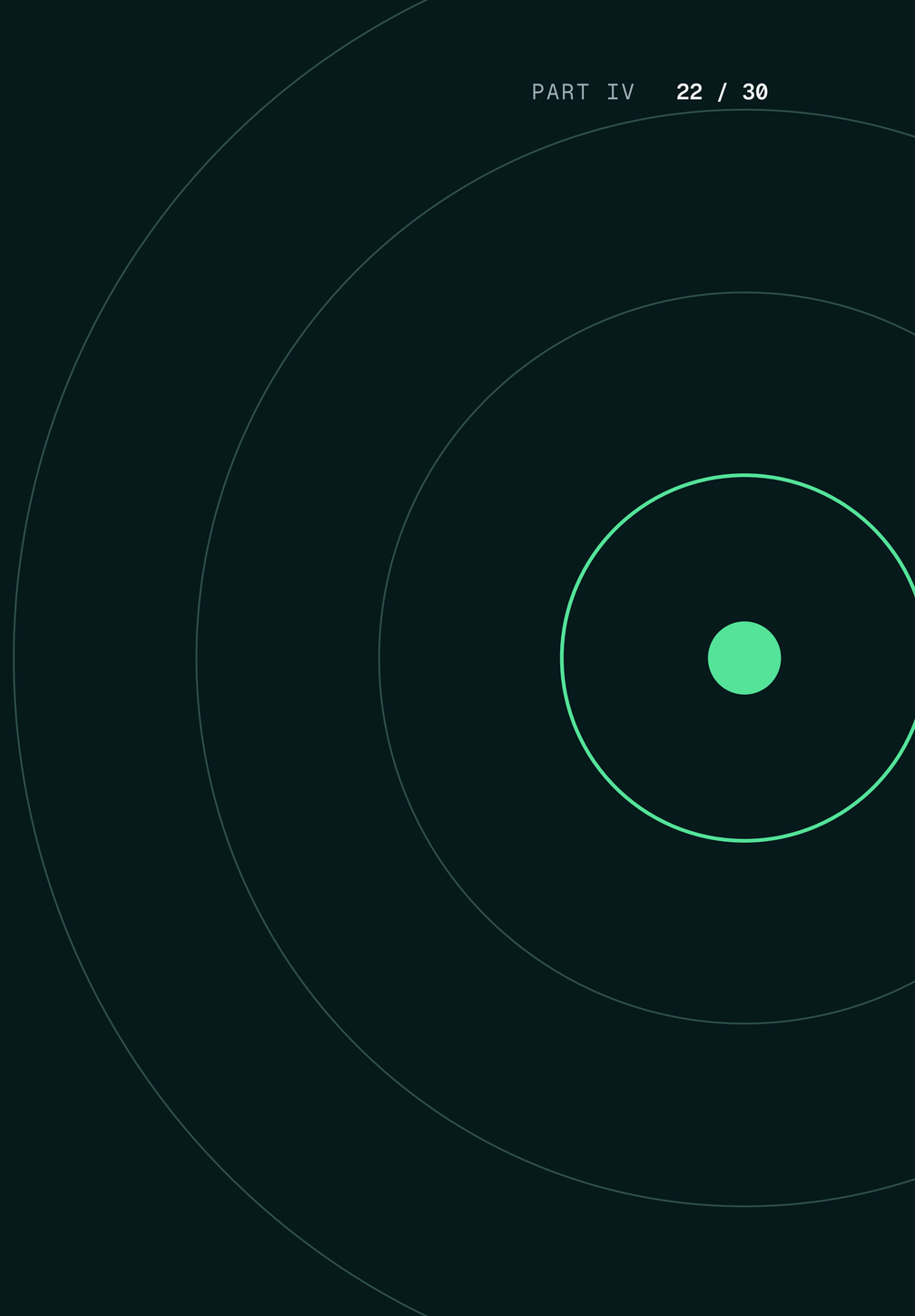
Model minus rule B, 1,000 client-level bootstrap resamples, 95% intervals. Share of resamples with the model ahead on hits: 77%, 87%, 66%. Each top-50 hit rate rests on 50 pages, so intervals are wide.

PART IV

The action

Turning the findings into a monthly list an editor can finish, with one suggested first step for every page.

23 The playbook
24 Does the list work
25 The confidence label
26 July's review list
27 Reach and drift



THE PLAYBOOK

Every listed page gets one suggested first step, checked in a fixed order.

HOW THE LIST IS BUILT

1. Rank every page with a real shortfall by its missed clicks
2. Keep the top 50, with at most 10 from any one website
3. Label each with the model's confidence
4. Give each one suggested first step

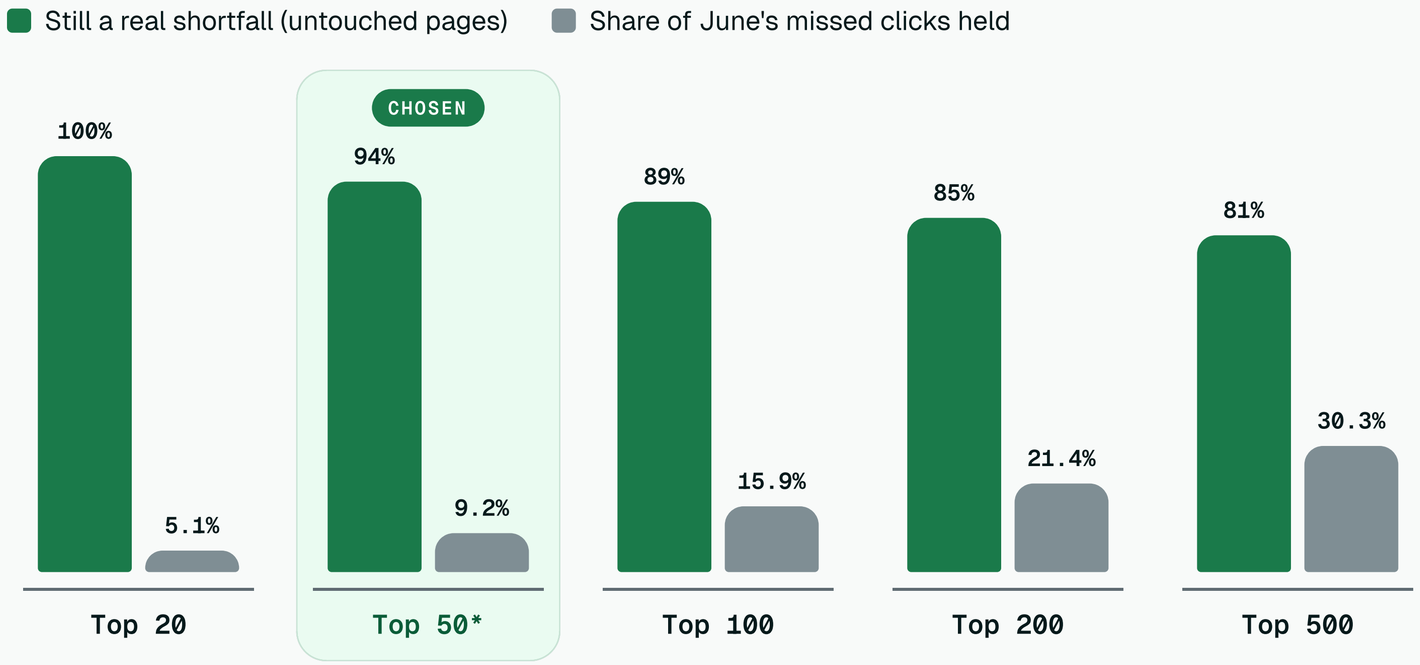
1	**Check indexing and rankings**	Impressions halved since last month: the page is leaving search, and a rewrite won't help.
2	**Wait and re-check next month**	Daily position swings by 5 or more places: these repeated least often (19%).
3	**Review the whole website**	The whole site is clicked at under half of what its positions predict.
4	**Check the results page first**	Buying-intent keywords, where ads and shopping results may take the clicks.
5	**Improve the content**	GA4 shows most visitors leave without engaging.
6	**Rewrite the title and description**	Everything else: the page is seen but its listing doesn't win the click.

UNDER THE HOOD

Reason codes: real_shortfall, client_wide_low_ctr, commercial_intent, low_engagement. Before acting, a person checks the live results page, that the page is still indexed, and whether it changed since the list was built. Nothing is published or de-indexed automatically.

DOES THE LIST WORK

In a back-test, 16 of the 17 listed pages that nobody touched were still losing clicks a month later.



94%

of untouched listed pages were real shortfalls again

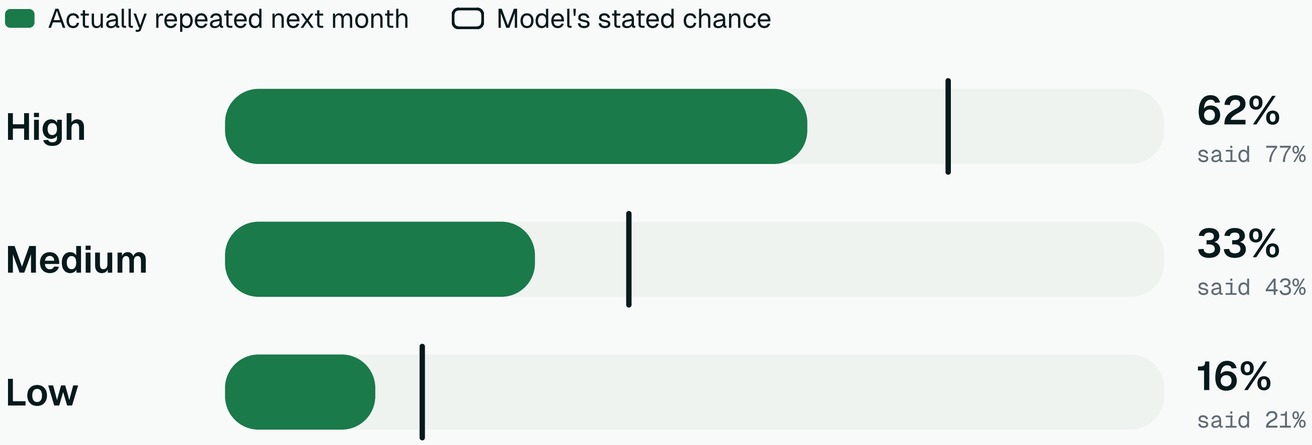
FlyRank's own team optimised 33 of the 50 during those two months. A longer list holds more of the missed clicks but hits less often; 50 keeps hits high.

UNDER THE HOOD

May → June back-test. *Chosen list: top 50 with at most 10 per website (94% hits among untouched pages, 9.2% of June's missed clicks, 11 websites); other sizes have no per-site cap. 0 pages vanished from search. Hit rates use only pages FlyRank did not optimise.

THE CONFIDENCE LABEL

The model's confidence label sorts pages in the right order, though its numbers run high.



About 4×

as many high-confidence pages repeated as low ones

Use the label to decide what to look at first, not as a promise. Its stated chances run 5 to 15 points above what happened.

UNDER THE HOOD

Whole queue, May → June back-test, pages FlyRank left alone: 919 high, 1,482 medium, 1,359 low. All 50 pages on the review list were labelled high (average stated chance 96%).

THE FIRST LIVE LIST

July's review list starts with pages missing thousands of clicks each, every one with a first step.

RANK	SUGGESTED FIRST STEP	CONFIDENCE	MISSED CLICKS, JUNE	IMPRESSIONS	POSITION	INTENT
1	**Review the whole website**	HIGH	**13,201**	585,712	2.2	Informational
2	**Check the results page first**	HIGH	**12,778**	292,416	1.9	Transactional
3	**Review the whole website**	HIGH	**12,598**	591,696	2.3	Informational
4	**Review the whole website**	HIGH	**6,702**	269,943	2.1	Informational
5	**Rewrite the title and description**	HIGH	**6,457**	157,289	1.8	Informational
6	**Check the results page first**	HIGH	**3,656**	116,908	1.9	Transactional
7	**Check the results page first**	HIGH	**3,359**	75,720	2.5	Transactional
8	**Review the whole website**	HIGH	**2,949**	119,228	2.4	Informational

UNDER THE HOOD

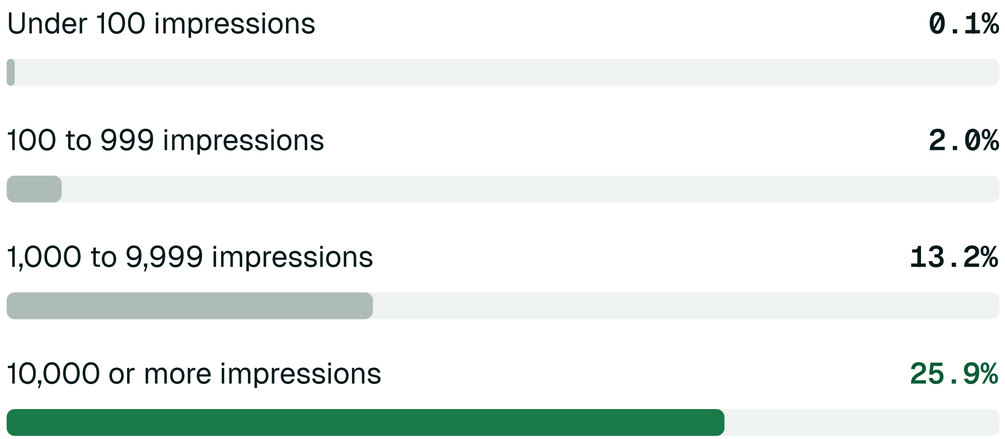
Built from June 2026 data; the first 8 of 50 are shown. Pages are identified only by pseudonymous hashes, left out here. All 50 are labelled high confidence, and 47 of them have 10,000 or more impressions.

REACH AND MONITORING

The list mostly reaches big pages, and the data has started to drift, so the model is due for retraining.

Who the list can reach

Share of pages in the queue, by June impressions. Small pages rarely have enough traffic to rule out luck.



Monthly monitoring

The list's hit rate held up, but the mix of positions shifted past its trigger.

MONTH PAIR	HIT RATE	POSITION PSI	STATUS
Feb → Mar	68%	0.000	Stable
Mar → Apr	80%	0.052	Stable
Apr → May	84%	0.148	Stable
May → Jun	88%	**0.561**	DRIFT
June (live)	–	**0.299**	DRIFT

UNDER THE HOOD

Drift = population stability index (PSI) on position against the first month; trigger at PSI ≥ 0.20. Impressions PSI reached 0.204 in May → June. Share of the model's misses that lost search visibility: 31%, 40%, 25%, 17%.

LIMITATIONS

What this work does not show yet.

No measured gain yet

It shows where clicks are missing and that the gap lasts, not how many clicks a fix wins back. Some gaps come from ads or AI answers no rewrite can beat.

Association, not cause

Every pattern here was observed in one portfolio over five months. None of it is a causal claim.

Fifty pages per test

Each top-50 hit rate rests on 50 pages, so intervals are wide and no single lead of the model is clear of chance.

One sealed month

It rules out tuning towards the test, but cannot show behaviour across seasons, and the mix of positions has already shifted since February.

Small pages are rarely judged

The list reaches a quarter of pages with 10,000+ impressions, but only 0.1% of the smallest.

Losing visibility is a different problem

Most of the model's misses were pages that dropped out of search the next month, which no click-based score can foresee.

UNDER THE HOOD

Changes after earlier runs: gradient boosting's stopping rule and rule B's two signals were added after reading earlier development runs. The comparison rule and the sealed test were fixed in advance.

WHAT'S NEXT

Next, measure the real gain with a fair experiment, and retrain now that drift has crossed its trigger.

01 **Run a randomised test of fixes**

Fix a random half of one month's list and compare it with the other half. That turns "where clicks are missing" into "how many clicks a fix wins back".

02 **Retrain and re-check the luck test**

Position drift crossed its trigger in June. Refit the model and re-estimate the spread factor on recent months.

03 **Add signals for pages leaving search**

Most misses lost visibility first. Flag those early, so they get an indexing check instead of a rewrite.

THE LESSON I'M KEEPING

> Give the simple approach a fair chance before trusting a complex one, and say so when it holds up.

Neil Ivan Orencia

THANK YOU

Find the pages that are losing clicks, then fix them.

Every number in this report traces back to the project notebooks, which hold the full method, code and saved outputs.

Neil Ivan Orencia
Machine Learning Intern, FlyRank

DATA CREDIT

Data provided by FlyRank (flyrank.ai) through the FlyRank ML Internship warehouse, release v20260703, pseudonymised before release. No client name, URL or search query appears in this work. Built with Python, DuckDB and scikit-learn on Google Colab.

